

Software Engineering Project

Jan 2026 Term - Team 051

Milestone - 5



Submitted by:

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1. Demo Video

Demo Video Link — [Link](#)

The demo video provides a complete walkthrough of the DineFlow Restaurant Management System across all four user roles — Manager, Chef, Waiter, and Customer. The video covers the full order lifecycle from a customer scanning a QR code at their table, placing an order, the manager approving it, the chef marking it prepared, the waiter delivering it, and the table being closed after payment. The AI-powered custom dish generation feature is also demonstrated live, showing a customer describing dietary preferences and receiving a generated recipe with cost and ingredients. The video is within the 5-minute limit.

2. Presentation (PPT)

Presentation Link — [Link](#)

The presentation provides a concise overview of the DineFlow project, structured around the core deliverables of user identification and user stories. The slides cover the team introduction, project objective, identification of all user roles (Manager, Chef, Waiter, Customer), the 11 user stories mapped to each role, the order lifecycle flow diagram, the tech stack, and key implementation highlights from Sprint 1 and Sprint 2.

Presentation covers:

- Team 051 introduction and project overview
- User identification: Primary (Manager), Secondary (Customer, Chef), Tertiary (Waiter)
- 11 user stories across all roles with SMART criteria
- Complete order lifecycle: Customer → Manager → Chef → Waiter → Billing
- Tech stack: Flask, SQLAlchemy, SQLite, Vue 3, Tailwind CSS, Chart.js, GenAI
- Sprint 1 and Sprint 2 key deliverables summary

3. Complete Project Code

Project Code Link — [Link](#)

The complete project source code, including the full Flask backend, all Vue 3 frontend components, the SQLite database, a consolidated test file with all 163 pytest test cases, and a detailed README.md with step-by-step instructions to set up and run the application, has been submitted as a ZIP file. The README covers prerequisites, virtual environment setup, dependency installation, environment variable configuration for AI providers, and how to start the server and access each role-specific dashboard.

4. Summary of Tools and Technologies

Category	Tool / Technology	Purpose
Backend	Python 3.11+	Primary programming language
Backend	Flask 3.0	REST API server and frontend file serving
Backend	Flask-SQLAlchemy 3.1	ORM for database access
Backend	Flask-CORS 4.0	Cross-origin request support for local dev
Backend	Werkzeug 3.0	Password hashing and request utilities
Backend	python-dotenv 1.0	Environment variable management (.env)
Backend	requests 2.31	HTTP client for AI provider API calls
Backend	PyJWT	JWT token generation for manager authentication
Database	SQLite	Lightweight embedded relational database
Frontend	Vue 3 (CDN)	Reactive UI component framework
Frontend	Tailwind CSS (CDN)	Utility-first CSS for layout and styling
Frontend	Bootstrap Icons (CDN)	Icon library for UI elements
Frontend	Chart.js (CDN)	Analytics charts on manager dashboard
Frontend	Google Fonts	Playfair Display + DM Sans typography
Frontend	HTML / CSS / JavaScript	Core web technologies (no build step)
AI Integration	Google Gemini (gemini-2.5-flash)	AI provider for custom dish generation
AI Integration	Anthropic Claude (claude-sonnet)	Alternative AI provider for custom dish generation
Testing	pytest	Test framework for all 163 API test cases
Project Mgmt	Jira (Kanban + Gantt)	Sprint planning, task tracking, timeline
Project Mgmt	GitHub	Version control and issue tracking
Documentation	OpenAPI 3.0 (Swagger YAML)	API documentation covering all 37+ endpoints
Development	VS Code + Codex extension	IDE with AI coding assistance

AI Usage Note

The AI integration in DineFlow supports two providers — Google Gemini and Anthropic Claude — switchable via the `AI_PROVIDER` environment variable. The AI service generates custom dish recipes based on customer dietary preferences, maps ingredients to existing inventory, normalises quantities to one serving, calculates pricing using the manager-defined profit margin, and generates chef-ready

cooking steps. Additionally, the Codex VS Code extension was used during development for backend coding assistance.

5. Issue Tracking Log

All issues were tracked on GitHub throughout the project lifecycle using labelled issues assigned to sprint milestones. A total of 24 issues were raised, tracked, and closed across all milestones. The table below lists all 24 issues. Screenshots of the GitHub Issues board are included below the table.

#	Issue Title	Labels	Status	Milestone
#24	Final submission — video demo, presentation, PDF report, README, code zip.	documentation	Closed	Milestone 5
#23	Update Swagger YAML + pytest for Sprint 2 + Submit Sprint 2 report.	documentation sprint-2 testing	Closed	Sprint 2
#22	DB changes for Sprint 2 — billing fields, selling_price, kitchen_status.	backend sprint-2	Closed	Sprint 2
#21	Sprint 2 - Add Items Sold card + build Billing Settings section (user feedback).	backend enhancement frontend sprint-2	Closed	Sprint 2
#20	Write Swagger YAML + pytest suite.	documentation sprint-1 testing	Closed	Sprint 1
#19	BUG: GET /inventory returns HTTP 200 without auth - fix @require_manager_token.	backend bug sprint-1	Closed	Sprint 1
#18	Implement all backend API endpoints - Manager, Customer, Chef and Waiter.	backend sprint-1 user Story	Closed	Sprint 1
#17	Submit Milestone 2 PDF — components, diagrams, frontend screenshots, frontend ZIP.	documentation frontend	Closed	Milestone 2
#16	Build Chef, Waiter and Customer frontend pages.	frontend	Closed	Milestone 2
#15	Build Manager frontend — Dashboard, Orders, Menu, Inventory, Analytics, Settings.	frontend	Closed	Milestone 2
#14	Scrum meeting minutes (6 meetings) + business owner feedback.	documentation	Closed	Milestone 2
#13	Set up Jira Kanban board, Gantt chart and sprint schedule.	documentation	Closed	Milestone 2
#12	Create class diagram and ER diagram.	backend design	Closed	Milestone 2

The remaining issues (#1–#11) were raised and closed during Milestone 1 and early Milestone 2, covering user research, user story writing, business owner interviews, SMART criteria definition, and initial project planning. All 24 issues across the project are now closed.

GitHub Issues Board — Screenshots

is:issue state:closed

🔍

🏷️ Labels

📅 Milestones

New issue

☐ Open **0** **Closed 20**

Author ▾ Labels ▾ Projects ▾ Milestones ▾ Assignees ▾ **Newest** ▾

☐ ☒ **Final submission — video demo, presentation, PDF report, README, code zip.** documentation milestone-5
#24 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 5 - Fin...

☐ ☒ **Update Swagger YAML + pytest for Sprint 2 + Submit Sprint 2 report.** documentation sprint-2 testing
#23 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 2 - API & ...

☐ ☒ **DB changes for Sprint 2 — billing fields, selling_price, kitchen_status.** backend sprint-2
#22 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 2 - API & ...

☐ ☒ **Sprint 2 - Add Items Sold card + build Billing Settings section (user feedback)** backend enhancement frontend sprint-2
#21 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 2 - API & ...

☐ ☒ **Write Swagger YAML + pytest suite.** documentation sprint-1 testing
#20 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 1 - API & ...

☐ ☒ **BUG: GET /inventory returns HTTP 200 without auth - fix @require_manager_token.** backend bug sprint-1
#19 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 1 - API & ...

☐ ☒ **Implement all backend API endpoints - Manager, Customer, Chef and Waiter.** backend sprint-1 user Story
#18 · by 23f1001130 was closed 35 minutes ago · 📌 Sprint 1 - API & ...

☐ ☒ **Submit Milestone 2 PDF — components, diagrams, frontend screenshots, frontend ZIP.** documentation frontend milestone-2
#17 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

☐ ☒ **Build Chef, Waiter and Customer frontend pages.** frontend milestone-2
#16 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

☐ ☒ **Build Manager frontend — Dashboard, Orders, Menu, Inventory, Analytics, Settings.** frontend milestone-2
#15 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

☐ ☒ **Scrum meeting minutes (6 meetings) + business owner feedback** documentation milestone-2
#14 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

☐ ☒ **Set up Jira Kanban board, Gantt chart and sprint schedule.** documentation milestone-2
#13 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

☐ ☒ **Create class diagram and ER diagram.** backend design milestone-2
#12 · by 23f1001130 was closed 35 minutes ago · 📌 Milestone 2 - Sc...

Final submission — video demo, presentation, PDF report, README, code zip. #24

Closed

23f1001130 opened 47 minutes ago

Video demo: full walkthrough of all 4 roles, complete order lifecycle. AI custom dish generation demonstrated live.

Presentation: 14 slides covering team, project overview, user identification, all 13 user stories, order lifecycle diagram, tech stack.

Final PDF: all milestones and sprints compiled, tools and technologies, issue tracking screenshots, code review and PR screenshots.

README: backend + frontend setup, environment variables (env for Gemini/Claude API keys), project structure.

Complete code zip submitted.

Create sub-issue

23f1001130 added this to the Milestone 5 - Final Submission milestone 47 minutes ago

23f1001130 assigned 23f1001130, 22f3001012, 23f1000687 and Nachiket-GORE 47 minutes ago

23f1001130 added this to DineFlow — Team 051 47 minutes ago

23f1001130 added documentation milestone-5 47 minutes ago

github-project-automation moved this to Backlog in DineFlow — Team 051 47 minutes ago

23f1001130 closed this as completed 36 minutes ago

github-project-automation moved this from Backlog to Done in DineFlow — Team 051 36 minutes ago

Assignees

22f3001012

23f1000687

23f1001130

Nachiket-GORE

Assign to Agent

Labels

documentation

milestone-5

Projects

DineFlow — Team 051

Status In progress

Milestone

Milestone 5 - Final Submission

Closed 36 minutes ago, 100% complete

Relationships

None yet

Development

Create a branch for this issue or link a pull request.

Notifications

Unsubscribe

You're receiving notifications because you're subscribed to this thread.

6. Final Comprehensive Report

This Milestone 5 document serves as part of the final comprehensive report for the DineFlow project. The complete report compiles all milestones and sprints — Milestone 1 (user research, user stories), Milestone 2 (frontend design, ER diagram, class diagram, Scrum meetings), Milestone 3 (Sprint 1 — backend APIs, Swagger YAML, pytest, user feedback), and Milestone 4 (Sprint 2 — user feedback implementation, updated APIs, updated test cases). The key deliverables across the full project lifecycle are summarised below.

Project Deliverables Summary

Milestone 1 — User Research & Stories

- Identified 4 user roles: Manager (primary), Customer and Chef (secondary), Waiter (tertiary)
- Conducted business owner interviews to gather pain points and requirements
- Wrote 11 SMART user stories covering all roles and workflows

Milestone 2 — Design & Frontend

- Built complete frontend: Manager dashboard (7 sections), Customer app, Chef kitchen station, Waiter dashboard

- Created ER diagram (9 tables) and UML class diagram
- Ran 6 Scrum meetings with documented minutes and business owner feedback sessions
- Set up Jira Kanban board, Gantt chart, and sprint schedule

Sprint 1 (Milestone 3) — Backend & Testing

- Implemented full Flask REST API: 37+ endpoints across 4 blueprints (Manager, Chef, Waiter, Customer)
- Written Swagger-compatible OpenAPI 3.0 YAML covering all endpoints with user story mapping
- Written pytest test suite with 163 test cases across 29 classes — covering positive, negative, and edge cases
- Collected user feedback: 2 feedback points (Items Sold card + Billing settings section)
- Identified and documented 1 bug: GET /inventory missing auth decorator — fixed with @require_manager_token

Sprint 2 (Milestone 4) — User Feedback Implementation

- Feedback #1: Verified Items Sold summary card working correctly on Analytics page
- Feedback #2: Added complete Billing Settings section — GSTIN, address, GST rate, bill_terms, payment QR, billing toggles, custom dish profit margin
- 12 new database columns added across Manager, Dishes, and Order Items tables
- Updated Swagger YAML and pytest test cases to reflect all Sprint 2 changes

Sprint Summary Table

Sprint	Period	Focus	Key Output
Sprint 1	Milestone 3	Backend API + Testing + Swagger + User Feedback	37+ endpoints, 163 tests, 1 bug fixed, Swagger YAML, feedback collected
Sprint 2	Milestone 4	User Feedback Implementation + Updated Docs	Billing settings, 12 DB columns, updated YAML and pytest, all 163 tests pass